

MAT1002: Calculus II L02

Quiz 3 — 2025-2026, Term 2

Wednesday, 8 Apr 2026

1. (a) (5 points) Show that the following limit does not exist,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}.$$

- (b) (5 points) Does

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2},$$

exist? If so, prove it; if not, justify it.

Solution:

- (a) Along the curve $y = kx$, $(x, y) \rightarrow 0$ as $y \rightarrow 0$, so

$$\lim_{(x,y) \rightarrow (0,0) \text{ along } y=kx} \frac{xy}{x^2 + y^2} = \lim_{y \rightarrow 0} \frac{xkx}{x^2 + (kx)^2} = \frac{k}{k^2 + 1} = \begin{cases} \frac{1}{2}, & \text{along } y = x, \\ -\frac{1}{2}, & \text{along } y = -x. \end{cases}$$

Hence, the limit does not exist.

- (b) On the x -axis, i.e., $y = 0, x \neq 0$, $f(x, y) = (x^2 - y^2)/(x^2 + y^2) \equiv 1$. On the y -axis, i.e., $x = 0, y \neq 0$, $f(x, y) \equiv -1$. Hence,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

does not exist.

2. (a) (6 points) Let $\hat{u} = \langle u_1, u_2 \rangle$ be any unit vector, find the directional derivative of $f(x, y)$ at $(0, 0)$ in the direction of $\hat{u}$ with

$$f(x, y) := \begin{cases} \frac{xy^2}{x^2 + y^4}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Is $f(x, y)$ differentiable at $(0, 0)$?

- (b) (4 points) Find an equation for the tangent line to the ellipse $x^2/8 + y^2/2 = 1$ at the point $(-2, 1)$.

Solution:

(a) We have

$$\begin{aligned} D_{\vec{u}}f(0,0) &= \lim_{h \rightarrow 0} \frac{f(hu_1, hu_2) - f(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 u_1 u_2^2}{h(h^2 u_1^2 + h^4 u_2^4)} \\ &= \lim_{h \rightarrow 0} \frac{u_1 u_2^2}{u_1^2 + h^2 u_2^4} \\ &= \begin{cases} \frac{u_2^2}{u_1}, & u_1 \neq 0, \\ 0, & u_1 = 0. \end{cases} \end{aligned}$$

The two variable $f(x, y)$ is not differentiable at $(0, 0)$.

(b) Let $f(x, y) = x^2/8 + y^2/2$, then $f_x = x/4$ and $f_y = y$. Therefore, $f_x(-2, 1) = -1/2$ and $f_y(-2, 1) = 1$. The tangent line is

$$-\frac{1}{2}(x+2) + (y-1) = 0,$$

or

$$y = x/2 + 2.$$

3. (10 points) Find the global maximum and global minimum of the function

$$f(x, y) := 2 + 2x + 4y - x^2 - y^2$$

defined on the triangular region in the first quadrant bounded by the lines $x = 0, y = 0$ and $9 - x$.

Solution: We have $f_x = 2 - 2x, f_y = 4 - 2y$. Solving $f_x = f_y = 0$ gives $(x, y) = (1, 2)$, which is in the interior of $\mathbb{D}$. The boundary is $l_1 \cup l_2 \cup l_3$, with

$$l_1 := \{(0, y) : 0 \leq y \leq 9\}; \quad l_2 := \{(x, 0) : 0 \leq x \leq 9\}; \quad l_3 := \{(x, 9 - x) : 0 \leq x \leq 9\}.$$

On l_3 , $f(x, y) = f(x, 9 - x) = -43 + 16x - 2x^2 = p(x)$. Solving $p'(x) = 0$ gives $x = 4$, so $(x, y) = (4, 5)$ is a candidate. At end points $x = 0$ and $x = 9$, we have $(0, 9)$ and $(9, 0)$ also being candidates. By doing similar arithmetic for l_1 and l_2 , we can find the following candidates:

$$l_1 : (0, 0), (0, 2), (0, 9); \quad l_2 : (0, 0), (1, 0), (9, 0).$$

Computing the values of f at all these candidates,

$$f(1, 2) = 7, f(0, 0) = 2, f(0, 2) = 6, f(0, 9) = -43, f(1, 0) = 3, f(9, 0) = -61, f(4, 5) = -11,$$

we find that the global maximum is $f(1, 2) = 7$ and the global minimum is $f(9, 0) = -61$.